

SAFAR RIAD

Industrial Electrical Engineer | Automation & Power Systems | AI/ML & Cybersecurity

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PROFESSIONAL PROFILE

Industrial Electrical Engineer (Master's, USTHB) with a solid foundation in electrical drives, industrial automation and power systems, complemented by growing expertise in artificial intelligence, machine learning and cybersecurity applied to modern infrastructure. Combines hands-on engineering experience — PLCs and electrical drives at Siemens DZ, MV/LV distribution networks at Sonalgaz — with applied machine learning research for smart grid security.

EDUCATION

Master's in Industrial Electrical Engineering

2024 – 2026

USTHB, Algiers

- Coursework: Electrical Drives, Industrial Automation & PLC Programming, Power Systems & Distribution Networks, Control Theory, Applied Machine Learning.
- Thesis: Detection of False Data Injection Attacks (FDIA) in Smart Grid SCADA Systems using Machine Learning — IEEE 118-bus benchmark.

Bachelor's in Electrical Engineering

2021 – 2024

USTHB, Algiers

- Project: Photovoltaic charging system for an electric bicycle (MATLAB/Simulink, P&O MPPT algorithm).

Baccalauréat, Experimental Sciences

2021

Lycée des Frères Mohamed et Ali, Zemmouri, Boumerdès

PROFESSIONAL EXPERIENCE

Engineering Intern — Siemens DZ, Algiers

Dec. 2024 – Jan. 2025

- Hands-on experience in industrial automation, PLCs and electrical drives; documented installation and commissioning processes on Siemens hardware and TIA Portal.
- Collaborated with senior engineers on troubleshooting and preventive maintenance of electrical drive systems.

Engineering Intern — Sonalgaz, Bab Ezzouar Agency, Algiers

Dec. 2023 (2 weeks)

- Participated in MV/LV distribution network operations, on-site technical inspections and compliance report documentation.

ACADEMIC PROJECTS & RESEARCH

Fault Detection and Classification in Induction Motors

2023

- Generated a custom fault dataset via MATLAB/Simulink; compared LSTM, BiLSTM, GRU and RNN models, reaching up to 90% accuracy for industrial fault diagnosis.

Photovoltaic Charging System for an Electric Bicycle — Bachelor's Project

2024

- Designed and simulated a photovoltaic battery charging system in MATLAB/Simulink with a P&O MPPT algorithm; validated through voltage, current and power waveform analysis under variable irradiance.

Modeling of the Induction Machine in MATLAB/Simulink

2024

- Modeled and simulated the dynamic behavior of an induction machine (IM) in MATLAB/Simulink.

Machine Learning-Based Cyberattack Detection in Smart Grids — Master's Thesis

2026

- Trained and compared RF, DT, SVM, XGBoost and MLP classifiers on the IEEE 118-bus dataset for multi-class cyberattack classification; achieved over 97% accuracy with Random Forest.

Neural Identification and Control of a PMSM with Fuzzy Logic using Artificial Neural Networks

2026

- Used artificial neural networks (ANN) to identify the dynamic model of a permanent magnet synchronous machine (PMSM) and to control its speed by replacing the classical regulator with a neural regulator; work structured in two parts: ANN-based identification of the PMSM, then neural control of the PMSM.

PUBLICATIONS

- Safar, R. — "Machine Learning-Based Detection of False Data Injection Attacks in Smart Grids," conference paper, based on Master's thesis work.

TECHNICAL SKILLS

Automation & Drives: PLC Programming, TIA Portal, Electrical Drives, Commissioning & Maintenance

Power Systems: MV/LV Distribution Networks, Grid Operations, SCADA Systems

Simulation & Modeling: MATLAB/Simulink, PSIM, COMSOL Multiphysics, MPLAB, XLPro Calcul

Programming: Python, C, MATLAB Scripting

Machine Learning & AI: Scikit-learn, TensorFlow/Keras, Random Forest, XGBoost, SVM, LSTM/BiLSTM/GRU

Cybersecurity: Industrial/Critical Systems Security, Smart Grid Attack Detection, IDS/IPS Concepts, Network Security

Languages: Arabic (Native) · French (Professional) · English (Professional, B2)

CERTIFICATIONS & ACTIVITIES

- CS50: Introduction to Cybersecurity — Harvard University / edX, 2024
- Arduino and C Programming — UC Irvine / Coursera, 2024
- CELEC Club — USTHB Electrical Engineering Students Association (2024); ongoing self-directed learning in Automation, AI/ML & Cybersecurity via Coursera, edX and scientific publications.